

PORTFOLIO ANALYSIS & RISK ASSESSMENT

A Case Study on a Five-Asset
Portfolio

Presented by

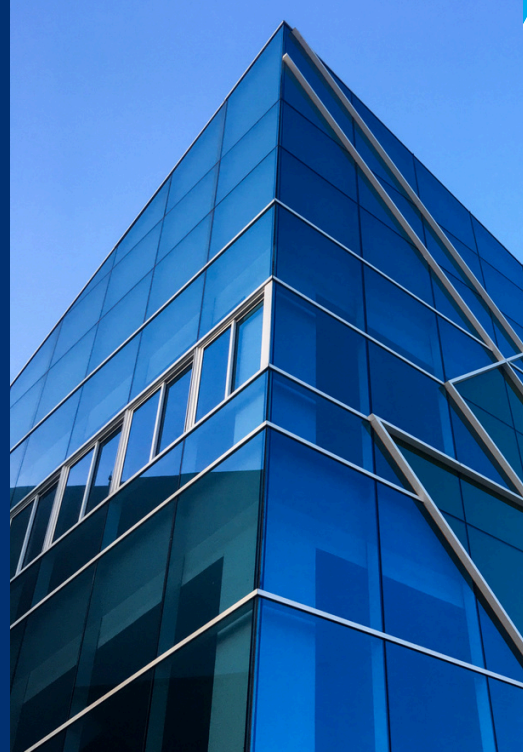
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INTRODUCTION & OBJECTIVE

This report analyzes a portfolio comprising five technology stocks: Tesla, Nvidia, Microsoft, Palantir, and Meta. The primary objective is to construct the portfolio, calculate its expected risk and return, and quantify the diversification benefits achieved by combining assets that are not perfectly correlated. Modern Portfolio Theory principles are applied to demonstrate the relationship between asset allocation and risk reduction.

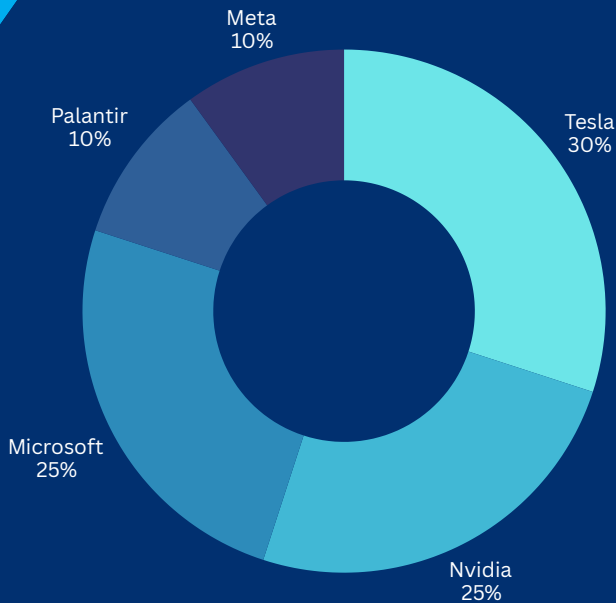


PROJECT FRAMEWORK

- **Data Collection & Preparation**
 - Gathered historical price data for five technology stocks.
 - Calculated daily returns and key statistical measures (variance, standard deviation).
- **Portfolio Construction**
 - Defined a target asset allocation strategy with specific weights.
- **Risk Modeling & Analysis**
 - Built a covariance matrix to quantify inter-asset relationships.
 - Calculated expected portfolio return, variance, and standard deviation.
 - Computed the theoretical portfolio risk assuming perfect correlation.
- **Performance Evaluation**
 - Quantified the diversification benefit by comparing actual vs. theoretical risk.
 - Prepared findings and recommendations based on Modern Portfolio Theory.

PORTFOLIO

This diversified allocation within the technology sector is engineered to capture growth while mitigating company-specific risk through deliberate weight distribution

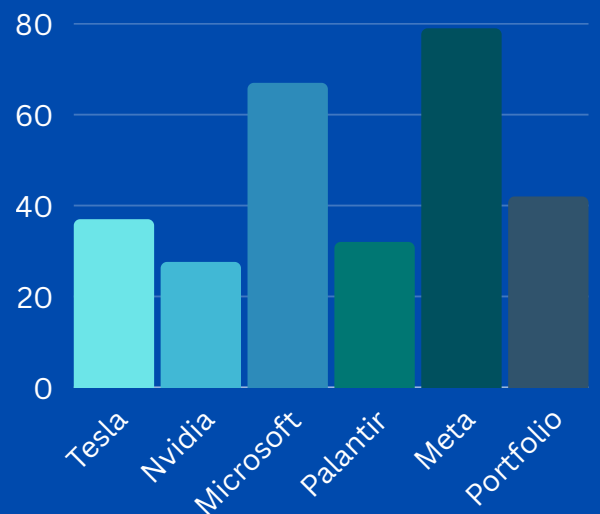


PORTFOLIO STRATEGY

This allocation reflects a focused strategy on the technology sector, with an emphasis on established industry leaders (Microsoft, Nvidia, Meta) balanced with targeted exposure to high-growth potential assets (Tesla, Palantir). The portfolio is designed to capture sector growth while managing risk through diversification across five key players

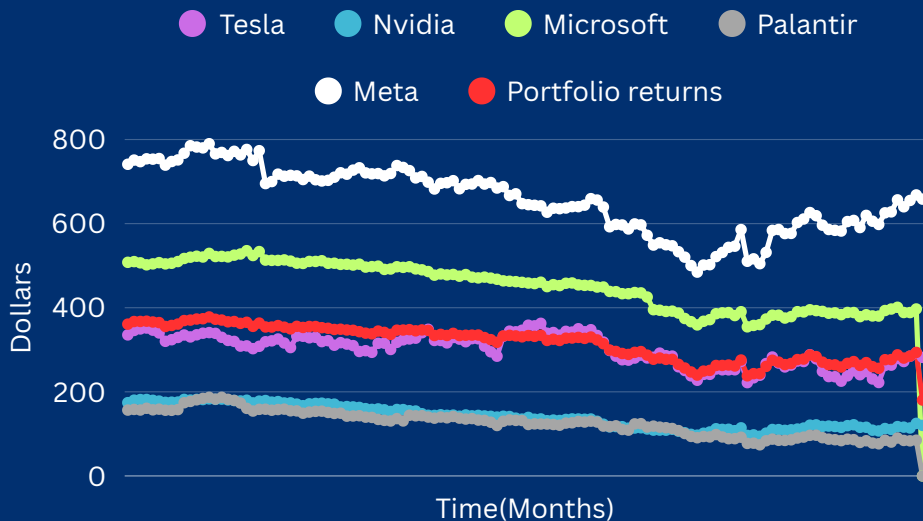
RISK COMPARISON

This comparison demonstrates the core benefit of our strategy: the constructed portfolio's volatility is significantly lower than the most volatile individual assets and below the average of its components



PORTFOLIO HIGHLIGHTS

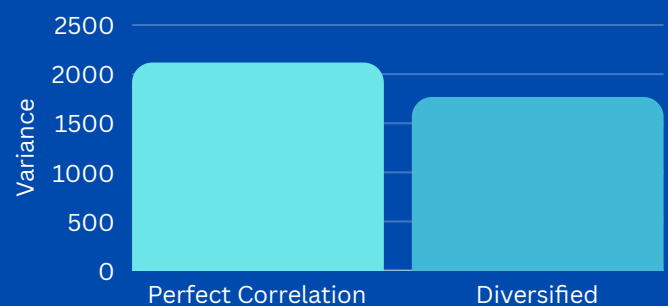
This analysis demonstrates the effective risk-return profile of the constructed portfolio. The portfolio achieved its primary objective of harnessing diversification, significantly reducing overall risk compared to a theoretical portfolio of perfectly correlated assets



STATISTICAL REPORT

- **Portfolio Expected Return:** 315.7\$
- **Portfolio Variance:** 1,767.34
- **Portfolio Standard Deviation (Volatility):** 42.04%
- **Diversification Benefit (Variance Reduction):** 367.0

Quantifying the Diversification Benefit



CONCLUSION & KEY FINDINGS

"This project successfully applied Modern Portfolio Theory to construct and analyze a five-asset technology portfolio. The key finding is the clear demonstration of diversification's power: the portfolio's variance (1,767.34) was significantly lower than its theoretical value if all assets were perfectly correlated (2,134.33).

This reduction in risk, achieved through a strategic 30/25/25/10/10 allocation, proves that combining assets with differing correlations is an effective method for optimizing a portfolio's risk-return profile.



WHAT'S NEXT?

Building on this analysis, several avenues for further research emerge:

- **Optimization:** Using Excel Solver to find the exact weights for the **Minimum Variance Portfolio** or the **Maximum Sharpe Ratio Portfolio** (Tangency Portfolio).
- **Benchmarking:** Comparing this portfolio's performance against a relevant benchmark, such as the Technology Select Sector SPDR Fund (XLK) or the S&P 500.
- **Expansion:** Incorporating additional asset classes (e.g., bonds, international stocks, commodities) to explore the benefits of broader diversification beyond a single sector

Reference

For the raw data and the calculations, scan QR code below

